WORKED SOLUTION

STEP II 1987, Question 14

Mechanics • Hooke's Law

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q14

The key idea

Because the band stays a horizontal circle on the cone, its whole state is described by one number: its distance s from the vertex along a generator. So we follow one small element. The inward pull from the tension has a component along the surface of the cone, which drives the element along the generator, and a component into the surface, which the smooth cone cancels. Converting from s to the length x gives simple harmonic motion, and the band goes slack after a quarter of a period.

Solution

The model. We treat the band as a thin elastic string of mass m spread uniformly round a circle. The cone is smooth, so its reaction on each element is normal to the surface, and we ignore gravity. By symmetry every element moves in the same way, so the band stays a circle centred on the axis. Let its distance from the vertex, measured along a generator, be $s(t)$. Its radius is then $s\sin\alpha$, because each generator makes angle α with the axis, and so

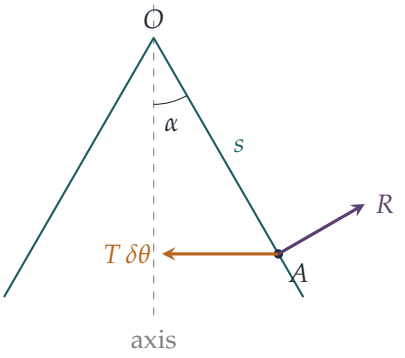
$$x = 2\pi s\sin\alpha.$$

While the band is stretched ($x > l$), Hooke's law gives tension $T = \dfrac{\lambda(x-l)}{l}$.

Forces on one element. Take an element subtending $\delta\theta$ at the axis. It is the fraction $\delta\theta/2\pi$ of the band, so its mass is $\dfrac{m\,\delta\theta}{2\pi}$. Two forces act on it:

- the given force $T\,\delta\theta$ from the tension, perpendicular to the axis and towards it;
- the normal reaction R from the cone, perpendicular to the generator through the element.

The diagram shows the plane containing the axis and the element.



The element is constrained to the surface, so it can move only along the generator OA. We resolve in that direction and perpendicular to it. The force $T\,\delta\theta$ is perpendicular to the axis, and the generator makes angle α with the axis, so the angle between $T\,\delta\theta$ and the direction $\overrightarrow{AO}$ is $90^\circ - \alpha$. Hence:

- along $\overrightarrow{AO}$ (towards the vertex), the component of $T\,\delta\theta$ is $T\,\delta\theta\cos(90^\circ - \alpha) = T\,\delta\theta\sin\alpha$, and R contributes nothing;
- perpendicular to the surface, $R = T\,\delta\theta\cos\alpha$, since there is no motion in that direction.

Since $R \geq 0$, the band really does stay pressed against the cone.

Ask yourself Why can't we just say the inward force $T\,\delta\theta$ makes the radius shrink?

The element can't move straight towards the axis: the cone is in the way. Only the part of $T\,\delta\theta$ along the surface does any accelerating. The rest is cancelled by the reaction. This is where the factor $\sin\alpha$ comes from, and it is the same reason a bead on a smooth slope feels only $mg\sin\theta$ along it.

The equation of motion. Newton's second law for the element along the generator, with s increasing away from the vertex, gives

$$\frac{m\,\delta\theta}{2\pi}\frac{\mathrm{d}^2 s}{\mathrm{d}t^2} = -T\,\delta\theta\sin\alpha \qquad \Longrightarrow \qquad \frac{\mathrm{d}^2 s}{\mathrm{d}t^2} = -\frac{2\pi T\sin\alpha}{m}.$$

The $\delta\theta$ cancels, so every element obeys the same equation, which confirms the band stays circular. Now $x = 2\pi s\sin\alpha$ with α constant, so

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} = 2\pi\sin\alpha\,\frac{\mathrm{d}^2 s}{\mathrm{d}t^2} = -\frac{4\pi^2 T\sin^2\alpha}{m} = -\frac{4\pi^2\lambda}{ml}(x-l)\sin^2\alpha,$$

using $T = \lambda(x-l)/l$. This holds while the band is taut, i.e. for $0 < t < t_0$, and rearranges to the required equation. □

Finding t_0. Put $y = x - l$ and $\omega^2 = \dfrac{4\pi^2\lambda\sin^2\alpha}{ml}$, so $\ddot{y} = -\omega^2 y$, which is simple harmonic motion about $y = 0$. Let $x_0 > l$ be the initial length (the band was pushed on,

so it starts stretched). The band is released from rest, so $\dot{y}(0) = 0$, and

$$x - l = (x_0 - l)\cos\omega t.$$

The band becomes slack when x first falls to l, that is, when $\cos\omega t = 0$ for the first time, so $\omega t_0 = \frac{\pi}{2}$. Since $\omega = 2\pi\sin\alpha\sqrt{\lambda/(ml)}$,

$$t_0 = \frac{\pi}{2\omega} = \frac{1}{4\sin\alpha}\sqrt{\frac{ml}{\lambda}}.$$

Notice that t_0 doesn't depend on how far the band was stretched, which is typical of simple harmonic motion: a quarter-period is the same whatever the amplitude.

Checks

- *Units:* λ is a force, so ml/λ has units of $\mathrm{kg\,m/N} = \mathrm{s}^2$, and t_0 is a time.
- *Limiting cases:* as $\alpha \to \frac{\pi}{2}$ the cone flattens into a plane, and the band contracts directly towards the centre with $t_0 = \frac{1}{4}\sqrt{ml/\lambda}$. As $\alpha \to 0$ the cone becomes a cylinder, the tension has no component along the surface, and $t_0 \to \infty$, as it should.
- *Numerical:* with $\alpha = 30°$, $m = 1$, $l = 1$ and $\lambda = 4$ the formula gives $t_0 = \frac{1}{4}$. Stepping the element's equation of motion numerically (and also computing the net tension force on a fine polygonal band directly in three dimensions) reproduces both the factor $\sin\alpha$ and t_0 to many decimal places for random choices of the parameters.

Common slips

- Forgetting the reaction of the cone and writing $\frac{m\,\delta\theta}{2\pi}\ddot{r} = -T\,\delta\theta$ for the radius r. This misses a factor of $\sin^2\alpha$ altogether.
- Using the mass m rather than $\frac{m\,\delta\theta}{2\pi}$ for the element.
- Giving the full period, or half of it, for t_0: the band starts at maximum stretch, so it reaches natural length after a quarter-period.

Beyond STEP

The given force $T\,\delta\theta$ is a small-angle version of a general fact: a string under tension T curving with radius of curvature ρ pushes sideways with force T/ρ per unit length. The same idea, extended to surfaces, explains why the pressure inside a soap bubble exceeds the pressure outside (Laplace's law). Motion constrained to a surface is handled most cleanly with Lagrangian mechanics, where the reaction never needs to appear. Kleppner and Kolenkow, *An Introduction to Mechanics*, has many problems on strings under tension and constrained motion.

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